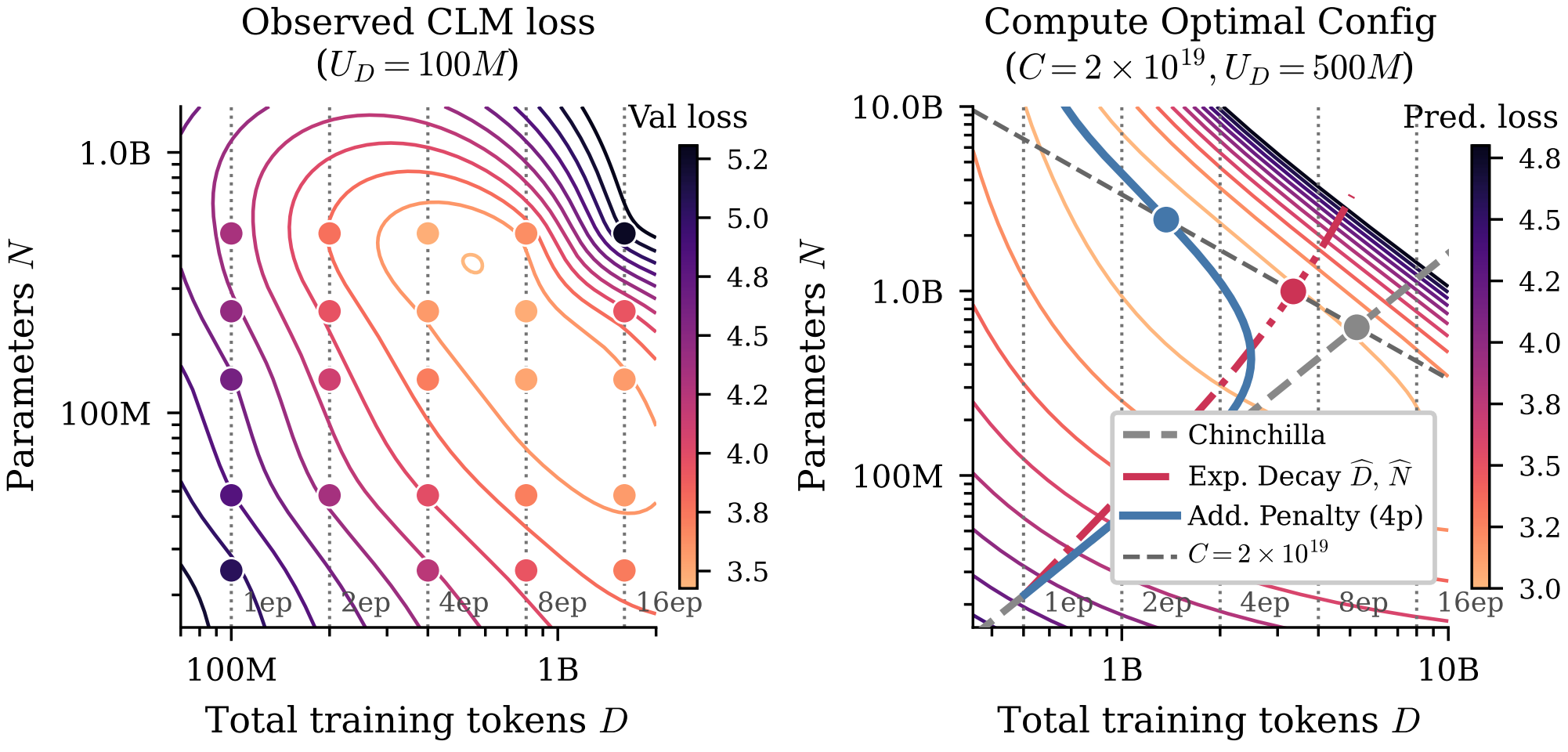
Observed CLM loss
($U_D = 100M$)
Val loss
5.2
5.0
4.8
4.5
4.2
4.0
3.8
3.5
1.0B
100M
Parameters N
1ep
2ep
4ep
8ep
16ep
100M
1B
Total training tokens D
Compute Optimal Config
($C = 2 \times 10^{19}, U_D = 500M$)
Pred. loss
4.8
4.5
4.2
4.0
3.8
3.5
3.2
3.0
10.0B
1.0B
100M
Parameters N
Chinchilla
Exp. Decay $\widehat{D}$, $\widehat{N}$
Add. Penalty (4p)
$C = 2 \times 10^{19}$
1ep
2ep
4ep
8ep
16ep
1B
10B
Total training tokens D